```
test@test-machine:~/$ ./pmq_sender_desc
Sender MQ /mq-postman opened with desc: 3
Sender: sending message: [Hello there - 0]
Sender: sending message: [Hello there - 1]
Sender: sending message: [Hello there - 2]
Sender: sending message: [Hello there - 3]
Sender: sending message: [Hello there - 4]
Sender: sending message: [Hello there - 5]
Sender: sending message: [Hello there - 6]
Sender: sending message: [Hello there - 7]
Sender: sending message: [Hello there - 8]
Sender: sending message: [Hello there - 9]
Sender: Exit
test@test-machine:~/$
```

Figure 1: A sender process opening a message queue and printing the descriptor

Run *pmq_receiver_desc* first in a terminal followed by *pmq_sender_desc* in another terminal.

The observation in Figure 1 and Figure 2 is that the *pmq_sender_desc* process opened the POSIX message queue *mq-postman* and got the descriptor as 3. *pmq_receiver_desc* process opened the same POSIX message queue as *pmq_sender_desc* process but got the descriptor as 4. This is because *pmq_receiver_desc* process had opened another POSIX message queue */mq-test*, which got descriptor 3 before opening the */mq-postman* queue. This shows that message send or receive works, although the descriptor values could possibly be different for different processes that are dealing with the same message queue.

Message queue full

The APIs *mq_send()* and *mq_receive()* return *EAGAIN* if the POSIX message queue is created in non-blocking mode (*O_NONBLOCK* set in *oflag*) and there is no message sent or received.

The API *mq_timedsend()* may block for a pre-set duration, if a queue is full and an attempt to send a message to this already full queue is made. Likewise, the API *mq_timedreceive()* may block for a pre-set duration, if the queue is empty and an attempt to receive a message from this already empty queue is made.

Single sender and multiple reader access

In an IPC, more than one process/program can open the same named POSIX message queue to write/read. Let us take an example where we have a single writer process and multiple reader processes. A single writer process opens a queue in *write* mode and the same queue is opened in *read* mode by more than one reader process. The multiple reader processes are nothing but the same program with multiple copies of its executable.

The code given below demonstrates the behaviour when multiple receiver processes are waiting to receive

```
test@test-machine:~/$ ./pmq_receiver_desc
Test MQ /mq-test opened with desc: 3
Receiver MQ /mq-postman opened with desc: 4
Receiver: blocked until message arrives...
Receiver: message received: [Hello there - 0]
Receiver: blocked until message arrives...
Receiver: message received: [Hello there - 1]
Receiver: blocked until message arrives...
Receiver: message received: [Hello there - 2]
Receiver: blocked until message arrives...
Receiver: message received: [Hello there - 3]
Receiver: blocked until message arrives...
Receiver: message received: [Hello there - 4]
Receiver: blocked until message arrives...
Receiver: message received: [Hello there - 5]
Receiver: blocked until message arrives...
Receiver: message received: [Hello there - 6]
Receiver: blocked until message arrives...
Receiver: message received: [Hello there - 7]
Receiver: blocked until message arrives...
Receiver: message received: [Hello there - 8]
Receiver: blocked until message arrives...
Receiver: message received: [Hello there - 9]
Receiver: Exit
test@test-machine:~/$
```

Figure 2: A receiver process opening a message queue and printing the descriptor

the messages from the same message queue. A single process sends a series of messages to the message queue.

```
// pmq_sender_ssmr.c

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <mqueue.h>

#define MQ_QUEUE_PERMISSIONS 0660
#define MQ_MAX_MESSAGES 10
#define MQ_MAX_MSG_SIZE 128
#define MQ_MSG_BUFFER_SIZE MQ_MAX_MSG_SIZE + 10

int main(int argc, char **argv)
{
        mqd_t mqd_sender;
        struct mq_attr attr;
        const char *mq_name = "/mq-postman";
        char tx_buffer[MQ_MSG_BUFFER_SIZE] = {"Hello there
-  "};

        attr.mq_flags = 0;
        attr.mq_maxmsg = MQ_MAX_MESSAGES;
        attr.mq_msgsize = MQ_MAX_MSG_SIZE;
        attr.mq_curmsgs = 0;

        if((mqd_sender = mq_open(mq_name, O_WRONLY | O_CREAT,
MQ_QUEUE_PERMISSIONS, &attr)) == (mqd_t)-1)
```